

Topological Feature Learning for Parameter Estimation in Neyman-Scott Processes

First-year progression report

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1 Introduction

The study of spatial point processes finds applications in astronomy, cell biology, epidemiology, etc (Møller and Waagepetersen, 2004). Many natural phenomena – such as galaxy clustering – are modelled using SPPs, so one can hope to leverage an understanding of these processes to obtain novel insights. The parameters of spatial point processes are of particular interest because they are interpretable scientific quantities. For example, when Neyman-Scott processes are used to model galaxy clustering the parent intensity κ corresponds to the number density of haloes hosting galaxies, the offspring intensity μ is the mean occupancy of a halo, and the cluster scale σ is the spatial extent of the galaxy distribution within a halo. In forest ecology, Waagepetersen and Guan (2009) fit Thomas processes to three tropical tree species and read the cluster scale directly as a summary of how tightly seeds are dispersed, finding that species dispersed by exploding capsules, by wind, and by birds and mammals separate along that parameter.

A number of parameter estimation methods have been proposed, typically relying on closed-form expressions for summary statistics such as the K and L functions. Minimum contrast estimation fits parameters by numerically minimizing a discrepancy between the empirical and theoretical curve, most commonly for K or g (Diggle, 2003). Palm likelihood instead builds a composite likelihood from the process’s Palm intensity, and has been developed specifically for Neyman-Scott processes (Tanaka et al., 2008). Fully Bayesian approaches treat the unobserved parent locations as latent variables and sample the posterior over θ by MCMC (Møller and Waagepetersen, 2004).

These methods share three limitations. First, they rely on closed-form summary statistics, which may not exist (the LGCP-Strauss process of Vihrs (2022) has no tractable density) or may exist but fail the regularity conditions the estimator’s theory needs (the two-step estimator of Waagepetersen and Guan (2009) requires K_ψ to be twice continuously differentiable in ψ , which excludes the Matérn cluster process). Second, they are sensitive to hyperparameters chosen per dataset by rule of thumb, with no held-out quantity indicating whether the choice was good: minimum contrast depends on $(r_{\max}, q, p)$, and Waagepetersen and Guan (2009) report that Thomas estimates of κ and ω are “quite sensitive” to $r_{\max}$ and q , being strongly negatively correlated so that the product $\hat{\kappa}\hat{\omega}$ is stable while the individual estimates are not. Third, they destabilise as the process approaches complete spatial randomness: writing $\lambda = \kappa\mu$ for the expected number of points per unit area and $\nu = \sigma\sqrt{\kappa}$ for the ratio of cluster radius to mean parent spacing, once $\nu \gtrsim 1$ neighbouring clusters overlap and many (κ, μ) pairs with the same product produce nearly the same observable pattern. The theoretical K -function makes this explicit – for a Thomas process $K(t) = \pi t^2 + (1 - e^{-t^2/4\sigma^2})/\kappa$, so as σ grows at fixed t the second term vanishes and K tends to the Poisson K regardless of κ – and the contrast surface is close to flat along this trade-off, so estimates become sensitive to the starting point.

This prompted me to consider whether persistence-based features could be used in this context, and whether they have advantages over these classical methods: they provide a multi-scale, process-agnostic descriptor of a point cloud that requires neither a closed form nor a choice of summary statistic.

1.1 Problem setup

Let X be a Neyman-Scott process on $\mathbb{R}^d$ with unknown parameter vector $\theta \in \mathbb{R}^{n_{\text{params}}}$. For example, the homogeneous Thomas process has $\theta = (\kappa, \mu, \sigma)$ corresponding to parent intensity, offspring intensity and cluster scale, respectively. We observe a single realization of X on a bounded window $W \subset \mathbb{R}^d$, giving a point cloud $\mathbf{x} = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$ with $n = |X \cap W|$. The objective is to produce an estimate $\hat{\theta} = \hat{\theta}(\mathbf{x})$ of θ from $\mathbf{x}$ alone. We stress *single realization*: θ is not recovered from repeated draws of X . This is the setting a practitioner faces: one observed pattern and one chance to estimate the generating parameters. It is also what makes the problem hard in a way no estimator can fix: a fixed θ already induces a distribution over point clouds, and it is this irreducible spread – not any weakness of a particular estimator – that sets an error floor.

The model is trained on a set of synthetic clouds $\{\mathbf{x}_i, \theta_i\}_{i=1}^{N_{\text{train}}}$ by backpropagating the RMS loss

$$\mathcal{L}(\hat{\theta}_i, \theta_i) = \sqrt{\sum_{j=1}^{n_{\text{params}}} (\hat{\theta}_i^j - \theta_i^j)^2}.$$

The scale of each parameter varies; therefore – if not normalized – the large-scale parameters will dominate this loss. To address this, we train in log-normalized parameter space, mapping $\theta_i \mapsto (\log(\theta_i) - \mu_{\log \theta}) / \sigma_{\log \theta}$, where – abusing notation – $\mu_{\log \theta}, \sigma_{\log \theta}$ denote the mean and standard deviation of the logged parameters over the training set. All losses reported below are in this space, so that $\mathcal{L} \approx 1$ corresponds to chance level.

2 Background

Vihrs (2022) first proposed the use of deep learning for parameter estimation. Yip et al. (2025) employ a similar architecture but use persistence images as the input feature, mapping H_0 – H_2 images of an α DTM ℓ filtration of a halo catalogue to (Ω_m, σ_8) ; their targets are cosmological parameters rather than the parameters of the underlying point process, and they do not systematically compare topological summaries. Which summary to use is a live question, since persistence diagrams do not live in a Hilbert space and cannot be fed directly to a machine learning model: Adams et al. (2017) propose persistence images, Umeda (2017) Betti curves, Bubenik (2015) persistence landscapes and Chazal et al. (2014) silhouettes, while Reininghaus et al. (2015) avoid vectorization altogether with a kernel on diagrams.

2.1 Neyman-Scott processes

A spatial point process on $W \subseteq \mathbb{R}^d$ is characterised by its intensity function $\rho(u)$, which satisfies $\mathbb{E}[N(W)] = \int_W \rho(u) du$ for $N(W) = |W \cap X|$; when ρ is constant the process is homogeneous, the canonical example being the Poisson process, whose points are scattered uniformly. A Neyman-Scott process is a clustered process built on top of one, and the processes used throughout this work are generated by the standard construction: expand the window W by an edge buffer to give W_{exp} , so that clusters with parents outside W are not truncated; sample a homogeneous Poisson *parent* process of intensity κ on W_{exp} ; for each parent p draw an offspring count $n_p \sim \text{Poisson}(\mu)$; place each offspring at $p + \epsilon$ with displacement ϵ drawn from a kernel k ; finally discard the parents and retain the offspring falling in W .

Every Neyman-Scott process therefore has the parent and offspring intensities (κ, μ) as parameters, and the choice of kernel supplies the rest. The only thing that changes between the families considered here is that kernel: for the *Thomas* process $k = \mathcal{N}(0, \sigma^2 I_d)$, giving $\theta = (\kappa, \mu, \sigma)$; for the Matérn cluster process k is uniform on a ball of radius R ; and for the *nested Thomas* process the parent process at the second step is itself a Thomas process rather than a Poisson one, producing clusters of clusters and a five-parameter target.

2.2 Persistent homology

Radially averaged statistics – such as the g and K functions – collapse a point cloud to a real-valued curve indexed by scale r , and are turned into features by evaluating over a range $[r_{\min}, r_{\max}]$ and subsampling to obtain a vector in $\mathbb{R}^n$. This discards multi-scale connectivity and void structure, i.e. *how* the density is distributed across space. Persistent homology (PH) instead tracks a growing family of simplicial complexes built on the cloud. A *filtration function* f assigns to each subset of points a scale at which it appears, subject to a monotonicity condition, and the sublevel sets $\mathcal{F}_r = \{\sigma : f(\sigma) \leq r\}$ form a nested family of complexes indexed by r . The canonical choice is Vietoris-Rips, where a simplex enters at its diameter; the DTM_k filtration (Anai et al., 2020) instead uses each point’s root-mean-square distance to its k nearest neighbours, so a simplex enters according to how densely its neighbourhood is occupied rather than how close its points happen to be.

As r grows, homological features of the complexes – connected components (H_0), loops (H_1), and higher-dimensional voids – are *born* and later *die* by merging into older features. Each feature is recorded as a pair (b, d) of birth and death scales, and the multiset of all such pairs in dimension h is the *persistence diagram* D_h ; the lifetime $d - b$ measures how robust a feature is, so points near the diagonal are usually read as noise. Applied to a point pattern, H_0 records the order and scales at which clusters merge, and H_1 records the gaps and voids the clustering leaves behind, neither of which survives radial averaging. Crucially, a persistence diagram requires no closed-form expression and no assumption about which family generated the pattern. Diagrams are multisets rather than vectors, so before they can be fed to a network they must be *vectorized*; the persistence image (Adams et al., 2017) does this by mapping to birth-persistence coordinates $(b, d - b)$, placing a weighted Gaussian bump at each point, and integrating the resulting surface over a 64×64 pixel grid.

3 ParamNet

The work completed so far consists of the simulator and its validation, the ParamNet model and its architectural ablations, comparison with alternative vectorizations, and benchmarking against existing baselines trained on the same data.

3.1 Synthetic cloud generation

Since the estimator is trained rather than derived, the simulator plays the role that a closed-form summary statistic plays in a classical method: it defines what the model is asked to learn. Training data are pairs $(\mathbf{x}_i, \theta_i)$ where θ_i is drawn from a *design distribution* over parameter space and $\mathbf{x}_i$ is a single realization of the corresponding process, generated by the construction of Section 2.1. The design distribution therefore acts as a prior, and the qualitative regimes it covers determine where the trained model is usable.

Sampling each parameter uniformly and independently from its own interval turns out to be inadequate, because the regimes of interest occupy curved subsets of the parameter grid and most of the mass would land on configurations that are near-Poisson or degenerate. I instead sample in reparameterized coordinates that carry the same meaning for every family:

the expected intensity $\lambda = \kappa\mu$, the parent intensity κ , and the dimensionless overlap index $\nu = r\sqrt{\kappa}$, where r is the cluster scale of the family in question ($r = \sigma$ for Thomas, $r = \sqrt{\sigma_1^2 + \sigma_2^2}$ for nested Thomas, with the extra coordinate $\rho = \sigma_2/\sigma_1$). Since ν measures cluster radius against mean inter-parent spacing, $\nu \ll 1$ gives well-separated clusters and $\nu \gtrsim 1$ clusters that merge. Draws are log-uniform, since the parameters span orders of magnitude, and are constrained so that $\lambda \in [150, 800]$ (persistence computation scales badly in the number of points), $\kappa \in [15, 120]$ (enough clusters per window to be informative, few enough to remain resolvable), and $\nu \in [0.10, 0.90]$. The upper limit on ν deliberately retains the near-Poisson regime in which classical estimators are least stable, rather than restricting to well-separated clusters. Rejection acts on the design and not on realizations, so it does not distort the conditional law of $\mathbf{x}$ given θ .

3.2 Method: ParamNet

Let $\mathbf{x}$ denote a point cloud, v its vectorized persistence summary, and $e \in \mathbb{R}^{E+1}$ the resulting feature vector, where E denotes the encoder’s embedding dimension and the $+1$ is the appended $\log N(\mathbf{x})$. ParamNet follows the pipeline

$$\mathbf{x} \xrightarrow{\text{Vectorization}} v \xrightarrow{\text{Encoder}} e \xrightarrow{\text{FCNN}} \hat{\theta}. \quad (1)$$

The pipeline shape follows the neural-estimator patterns used in Vihrs (2022) and Yip et al. (2025); what I have added is the application to a more challenging process (nested Thomas), an explicit evaluation of the design choices at each stage, and a comparison of TDA-based vectorizations, on which there has been no formalized discussion in this setting.

Filtration. My starting point was the canonical Rips filtration, but it is built from inter-point distances alone: it is sensitive to the sparsest connection between two regions and does not register how densely those regions are occupied. Since the parameters of a point process act on local density, DTM_k seemed the more natural object, and it is what the experiments bore out – Rips is the worst filtration arm on both processes, by a wide margin on nested Thomas (Table 1). Within DTM, error grows monotonically with k , consistent with a finer k -nearest neighbourhood retaining more of the local-density signal while a coarser one averages it away. I also tried fusing several scales to give the encoder fine and coarse views simultaneously; this helps on nested Thomas but not on Thomas, so the reported models use $k = 5$ for Thomas and fused $k \in \{5, 10, 15\}$ for nested Thomas. That this is a per-process choice is a weakness I return to in Section 4.

Encoder, head, and ablations. For Thomas I compute H_0 and H_1 persistence images of the DTM_5 diagram; for nested Thomas, H_0 and H_1 images for each of the three $\text{DTM}_{\{5,10,15\}}$ filtrations. All images are 64×64 . Each is passed through a small convolutional stack: three 3×3 blocks with channel widths (32, 64, 128), each followed by a ReLU and 2×2 max-pooling with spatial dropout ($p = 0.2$) between blocks. Each block is a CoordConv layer (Liu et al., 2018), concatenating two channels holding each pixel’s normalized coordinates before the convolution: a persistence image’s axes are not arbitrary – birth encodes local density and persistence encodes separation – so a pixel’s absolute location, not just its neighbourhood, is informative. The feature map is reduced by adaptive average pooling and projected to $E = 64$, $\log N(\mathbf{x})$ is appended, and the result is passed to an FCNN head with two hidden layers of width 64 and 32 (ReLU, dropout $p = 0.1$). When three scales are used they share one encoder and their embeddings are concatenated. Every experiment uses AdamW (lr = 10^{-3} , weight decay 3×10^{-4}), batch size 32, up to 500 epochs with early-stopping patience 50.

Each of these choices was ablated one axis at a time against this configuration (Table 1). A shared encoder beats three independent ones. H_0 alone is competitive while H_1 alone collapses

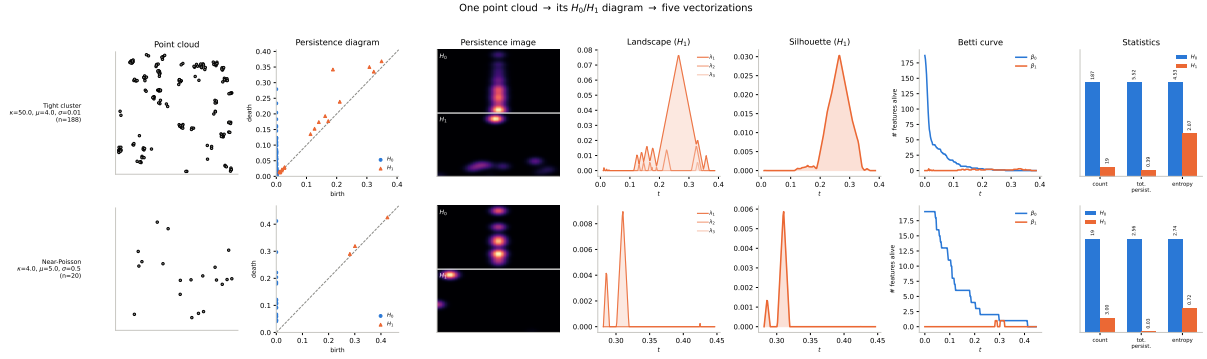


Figure 1: One point cloud $\rightarrow$ its H_0/H_1 persistence diagrams $\rightarrow$ the vectorizations compared in Table 1: persistence-image raster, landscape stack, silhouette curve, Betti curve, and persistence-statistics summary. Top row: the tight-cluster design ($\kappa = 50, \mu = 4, \sigma = 0.01$); bottom row: the near-Poisson design ($\kappa = 4, \mu = 5, \sigma = 0.5$). Statistics-panel bars are min-max normalized per statistic, with raw values printed above each bar, since count, total persistence, and entropy live on incomparable scales.

Axis	Arm	Thomas $\mathcal{L}$	Nested-Thomas $\mathcal{L}$
Filtration	Rips (vec_multik, landscape)	0.178 ± 0.008	0.461 ± 0.016
	DTM ₅ / DTM ₁₅	0.125 ± 0.007 / 0.146 ± 0.006	0.205 ± 0.003 / 0.260 ± 0.017
	DTM _{5,10,15} (fused)	0.126 ± 0.008	0.192 ± 0.010^c
Homology	H_0 only / H_1 only	0.125 ± 0.006 / 0.900 ± 0.019	$0.280 \pm 0.250^\dagger$ / 0.932 ± 0.028
Encoder	shared / independent, concat	0.125 ± 0.007 / 0.134 ± 0.009	0.201 ± 0.010 / 0.222 ± 0.021
Vectorization	Persistence image	0.126 ± 0.008	0.192 ± 0.010^c
	Landscape	0.132 ± 0.008	0.250 ± 0.005
	Silhouette ($p = 1$)	0.165 ± 0.013	0.271 ± 0.012
	Betti curve	0.196 ± 0.014	0.290 ± 0.005
Features	log N removed	$0.217 \pm 0.278^\dagger$	0.203 ± 0.007
	Topology removed (log N only)	0.893 ± 0.015	0.948 ± 0.019

Table 1: Branch summary: every axis measured against the multi-scale, H_0+H_1 , CoordConv starting configuration, $n = 10$ seeds per arm unless noted. Vectorization rows are the native-encoder path. A dagger (†) marks an arm on which at least one seed failed to converge and collapsed to the chance level, so the mean is dominated by the failed seed(s). $^cn = 3$. The DTM₅ / fused rows support the process-specific scale choice rather than ablating a single model.

to chance, so the merge structure carries most of the signal. Of the vectorizations, Figure 1 shows the four compared on a common diagram, and none of landscapes, silhouettes or Betti curves improved on persistence images on either process. Two further arms isolate where the signal lives: training on log N alone collapses to the chance level on both processes, while removing log N from the full model costs little on nested Thomas ($0.192 \rightarrow 0.203$) and is merely noisy on Thomas. The persistence images, not the point count, are doing the work.

3.3 Estimation performance and baseline comparison

I compare ParamNet against the classical and neural baselines of Section 1: minimum contrast fit to Ripley’s K and, separately, to the pair correlation function g , and vihrs – my re-implementation of the neural estimator of Vihrs (2022): a 1-D CNN over the edge-corrected, centred Ripley $L(r) - r$ curve plus $\log N(x)$, i.e. the pipeline of Section 3.2 with the persistence-image feature replaced by a single classical radial summary. All methods are run on the same clouds and splits, which are made by parameter vector so that no parameter value appears in both training and evaluation. Chance level is measured by retraining the default model on randomly permuted labels.

vihrs beats ParamNet on Thomas: its per-parameter losses ($\mathcal{L}^\kappa = 0.192$, $\mathcal{L}^\mu = 0.122$, $\mathcal{L}^\sigma = 0.029$) are each individually lower than ParamNet’s own ($0.201, 0.134, 0.040$; Table 3). So

Method	Thomas $\mathcal{L}$	Nested-Thomas $\mathcal{L}$
ParamNet [‡]	0.125 ± 0.007	0.192 ± 0.010
vihrs ($L(r) - r$ CNN)	0.115 ± 0.007	0.217 ± 0.006
Minimum contrast, K	0.855 ± 0.682^a	— ^d
Minimum contrast, g	6.199 ± 0.339^b	— ^d
Chance-level reference (label shuffle)	1.003 ± 0.019	1.006 ± 0.020

Table 2: Estimator comparison on Thomas and Nested Thomas, $n = 10$ seeds, on the shared test split. [‡]ParamNet is process-specific: DTM₅ for Thomas, fused DTM_{5,10,15} for nested Thomas. ^a Minimum contrast on K is unstable across seeds rather than uniformly bad, and ^b minimum contrast on g is consistently worse than chance; both are discussed in the text. ^dNot run: no closed-form summary statistic for nested Thomas has yet been derived, so its absence is an outstanding experiment rather than an impossibility.

(a) Thomas (κ parent intensity, μ offspring intensity, σ cluster scale)				
Parameter	ParamNet	vihrs	Min. contrast, K	Min. contrast, g
$\mathcal{L}^\kappa$	0.201 ± 0.010	0.192 ± 0.013	1.373 ± 1.087	8.825 ± 0.485
$\mathcal{L}^\mu$	0.134 ± 0.007	0.122 ± 0.007	0.939 ± 0.758	6.334 ± 0.370
$\mathcal{L}^\sigma$	0.040 ± 0.004	0.029 ± 0.002	0.254 ± 0.203	3.437 ± 0.170

(b) Nested Thomas		
Parameter	ParamNet ^c	vihrs
parent intensity	0.283 ± 0.023	0.320 ± 0.010
meta-offspring	0.402 ± 0.022	0.493 ± 0.022
meta-cluster scale	0.143 ± 0.006	0.132 ± 0.009
mean offspring	0.086 ± 0.004	0.110 ± 0.010
cluster scale	0.044 ± 0.000	0.028 ± 0.003

Table 3: Per-parameter loss $\mathcal{L}^j$ underlying the aggregate $\mathcal{L}$ of Table 2 (same methods, same seeds; ParamNet/nested is $n=3$). Minimum contrast has no nested-Thomas columns for the reason given there.

on this single-level process, the topological pipeline does not beat a hand-built radial summary. On nested Thomas, ParamNet gains an advantage on the density and count targets while vihrs maintains an edge on both scale parameters. $L(r) - r$ is built to resolve a characteristic radius directly, while a density-based topological feature is not, and it is only on the more complex two-level process that this trade-off favours the topological pipeline.

Both neural methods dominate minimum contrast on Thomas by roughly an order of magnitude, but for two different reasons. On K the estimator achieves losses of 0.365–0.399 for 6/10 seeds – close to vihrs – and 0.903–2.056 on the remaining 4/10, which is the near-Poisson flatness described in Section 1. On g every seed is worse than chance, indicating systematic bias rather than scattered non-convergence; the likely cause is my contrast exponent $c = 1$, set by analogy with Diggle (2003)’s K -based rule of thumb rather than tuned, where $c = 1/4$ is the value validated for aggregated patterns. I report the number but do not read it as a statement about contrast estimation on g in general.

Finally, parent intensity κ is the hardest target and the cluster scale σ the easiest, for every method on both processes and by a wide margin. This ordering holds regardless of estimator, so it looks like a property of the estimation problem itself rather than an artifact of any one feature set or architecture.

4 Discussion, limitations and future work

The results support three claims. Persistence images are informative about Neyman-Scott parameters: both processes sit roughly an order of magnitude below the label-shuffle ceiling, and the two “Features” arms of Table 1 place that signal in the images rather than in $\log N$. They are not better than a well-chosen radial summary on a single-level process: `vihrs` wins on all three Thomas parameters, and on both scale parameters of nested Thomas. The advantage is confined to the two-level process and to its density and count targets, which is what one would expect of a feature built from local density rather than from a characteristic radius. Two secondary observations: the ordering of parameter difficulty is stable across feature sets and architectures, so it reads as a property of the estimation problem; and the margin over minimum contrast is not a single effect, being seed-to-seed instability on K and a systematic bias traceable to an untuned contrast exponent on g . Neither number is a verdict on contrast estimation.

Limitations.

- **Multiple models.** The number of filtration scales is a per-process choice with no established rule beyond experimentation and comparison of validation losses, which undercuts the process-agnostic framing of Section 1. Always fusing would restore a single model at $3\times$ the persistence cost and 0.001 on Thomas, i.e. inside seed noise; attention over scales might be the fix.
- **Competitiveness.** Three alternative vectorizations were implemented and none improved on persistence images. This is evidence about these summaries under this architecture and this design distribution, not a general claim about vectorizations.
- **No advantage on Thomas.** `vihrs` wins on every parameter of the single-level process, and on both scale parameters of the nested one.
- **The design distribution is a prior.** The trained model approximates a posterior summary under it, so its ranges must cover the intended application domain.
- **Computational cost.** Persistence computation scales badly in n ; this is why the point budget of Section 3.1 is capped at 800. The inference cost per cloud is currently unmeasured.

Planned directions.

- The encoder architecture: aim to design a universal architecture that is process-agnostic.
- Inhomogeneous / non-stationary processes, where classical closed forms are least available and the topological approach has the most to gain. The comparator is the two-step procedure of Waagepetersen and Guan (2009), which needs the covariates observed and the intensity correctly specified; a simulation-trained estimator needs neither.
- Apply to real data. Galaxy catalogues if the running example of Section 1 is cosmological. The Barro Colorado Island tree data of Waagepetersen and Guan (2009) might be a second target: a small public data set already fitted with a Thomas process.

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